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Peatland restoration as an affordable nature-based climate solution with fire reduction and conservation co-benefits in Indonesia

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Peatland restoration as an affordable nature-based climate
solution with fire reduction and conservation co-benefits in
IndonesiaZ D Tan^{1,4,*}, L R Carrasco², S Sutikno³ and D Taylor¹¹ Department of Geography, National University of Singapore, Singapore 117570, Singapore² Department of Biological Sciences, National University of Singapore, Singapore 117558, Singapore³ Civil Engineering Department, University of Riau, Pekanbaru, Riau 28293, Indonesia⁴ Centre for Nature-based Climate Solutions, National University of Singapore, Singapore 117546, Singapore

* Author to whom any correspondence should be addressed.

E-mail: tzdienle@nus.edu.sg**Keywords:** nature-based solution, peatland, restoration, Indonesia, climate change, fire risk reduction, species conservationSupplementary material for this article is available [online](#)

Abstract

Ecosystem restoration is increasingly employed as a nature-based solution to a range of crises. Decisions over restoration must balance limited resources, land constraints, and competing demands. Peatlands in Southeast Asia have been heavily impacted by agricultural expansion over the past three decades, with Indonesia now accounting for a substantial proportion of degraded tropical peatlands globally. Using spatial linear programming, we focus on prioritizing peatland restoration sites in Indonesia for fire risk reduction, climate change mitigation, species conservation, and cost-effectiveness. The study finds that restoring peatlands at 1 km² planning units can generate multiple co-benefits such as reduced fire risks by 6%–37%, attenuated extinction risks of peatland specialist bird species and mitigated climate change potential of 0.002–0.36 Pg CO₂e yr^{−1}. These benefits were reduced but still of comparable magnitude when larger areas of planning (defined by village and catchment boundaries) were used. The results, although indicative, support tropical peatland restoration as a cost-efficient strategy for mitigating climate change, reducing fire, conserving biodiversity, and supporting sustainable development that can be offset by carbon prices of USD 2–37/Mg CO₂e.

1. Introduction

Recent decades have witnessed a rise in ecosystem restoration initiatives aimed at mitigating ecological degradation. Global estimates of degraded lands range from 1 to 6 billion hectare (Gibbs and Salmon 2015), with degradation concentrated in the biodiversity-rich tropics (Watts *et al* 2019). Initiatives such as the Post-2020 Global Biodiversity Framework demonstrate the urgent need to avoid irreversible damage to ecosystems and subsequent deterioration of social wellbeing, while pushing for more ambitious environmental goals and expanding markets for carbon and other ecosystem services (Koh *et al* 2021).

Over 140 Mha of tropical lands have been committed for restoration, with a median of 2 Mha

per commitment, mostly in developing countries (Brancalion *et al* 2019, Fagan *et al* 2020) characterized by rapid land cover change, pressing socio-economic and environmental challenges, and limited financial resources (Laurance *et al* 2014). Prioritizing restoration to deliver valuable ecosystem services can help build support for and alleviate constraints on restoration in economically marginalized countries while simultaneously contributing to (inter)national initiatives. Additionally, the various competing demands on land, e.g. for biomass production, biodiversity conservation, and climate change mitigation, present potential trade-offs. Evaluating and comparing the costs and benefits of each potential land use provides a means of selecting the most appropriate sites for restoration.

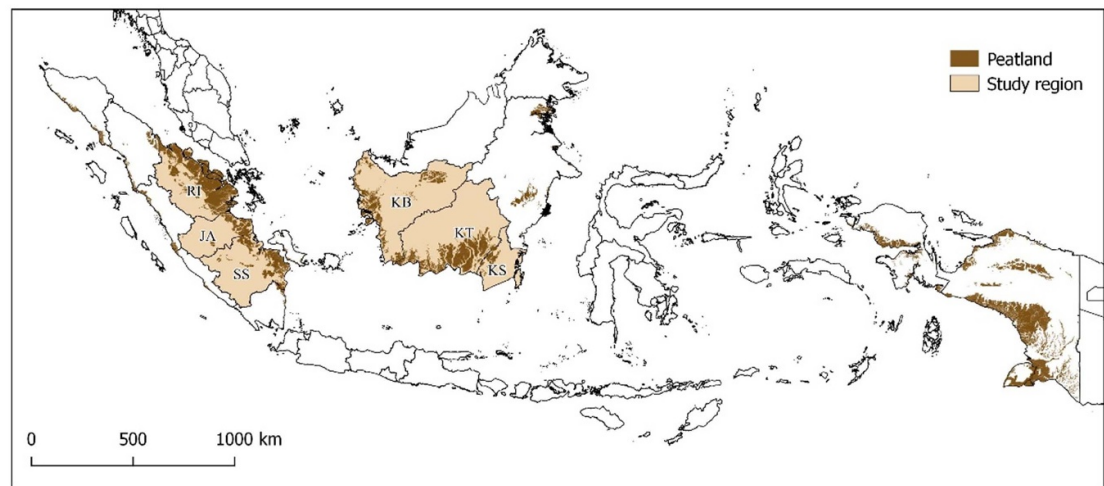


Figure 1. Peatland distribution across Indonesia and the six provinces in this study. RI = Riau; JA = Jambi; SS = South Sumatra; KB = West Kalimantan; KT = Central Kalimantan; KS = South Kalimantan.

Indonesia accounts for around 36% of tropical and 80% of Southeast Asian peatlands (Evers *et al* 2016, Warren *et al* 2017b). Peatlands in Indonesia are mainly found in several provinces distributed across the islands of Sumatra, Borneo (Kalimantan), and Papua (figure 1). Commencing in the 1990s, extensive areas of peatland were drained and the forests they supported were harvested to facilitate industrial plantations and smallholder-based agriculture expansion (Miettinen *et al* 2016b). By 2015, more than 70% of peatlands in Sumatra and Kalimantan had been damaged (Miettinen *et al* 2016b) and the ecosystem services and benefits they provide in their intact state severely compromised. In addition, degraded peatlands are prone to burning and constitute the primary source of transboundary haze (fire-related air pollution) that periodically blights the region, which has become a major source of geopolitical tension (Marlier *et al* 2019). Haze presents a major health concern, and contribute to excess morbidity and premature mortality in the region (Koplit *et al* 2016).

In the wake of a major haze episode in 2015, the Government of Indonesia established an ad hoc agency—the Peatland Restoration Agency (*Badan Restorasi Gambut*, BRG)—to oversee and facilitate the restoration of at least 2.5 Mha of peatlands by 2020 (BRG 2016), with an additional 1.2 Mha peatlands and 600 000 ha mangroves by 2024 under the new agency title Peatland and Mangrove Restoration Agency (BRGM). As BRGM represents a relatively new endeavor, this study focuses on BRG's restoration framework, comprising the rewetting of peatlands to restore water table levels, revegetation of native peat swamp forest species, and the revitalization of peatland community livelihoods (Dohong *et al* 2018). A fourth activity, fire mitigation, was implicit in the three other restoration activities.

Spatial optimization has been used in Indonesia to determine the area for restoration needed to offset the impacts of harmful land uses (Budiharta *et al* 2016, 2018) and to target specific peatland restoration interventions, such as the blocking of drainage canals and the prevention of fire (Santika *et al* 2020, Urzainki *et al* 2020). Kalimantan has featured prominently in these studies (Budiharta *et al* 2018, Santika *et al* 2020), which have tended to focus on a singular objective (Marlier *et al* 2019, Santika *et al* 2020), paying limited attention to the economic costs of restoration. At the national level, identification of priority sites for restoration was based on past histories of fires, topography, canal infrastructures, and the functional zoning of peatlands (BRG 2016). While this method accounts for restoration feasibility, the fact that potential benefits are not explicitly considered led to a lack of clearly defined goals and metrics to evaluate restoration outcomes (Puspitaloka *et al* 2020).

This study builds upon an optimization of tropical forest restoration for carbon sequestration and biodiversity conservation (Budiharta *et al* 2014, 2018) by expanding the geographical scope of interest to the rest of Indonesia and including an additional fire risk reduction objective. Additionally, multiple objectives for restoration are considered rather than single objectives such as fire mitigation (Marlier *et al* 2019, Santika *et al* 2020) and canal block allocation (Urzainki *et al* 2020) in isolation, while potential trade-offs and synergies in restoration outcomes are identified. The study determines the benefits and financial costs associated with restoring degraded tropical peatlands in Indonesia, and where restoration sites may be allocated to maximize multiple co-benefits while accounting for trade-offs and cost-effectiveness. The multiple objectives considered comprise: (a) species conservation,

(b) climate change mitigation, (c) fire risk reduction, (d) cost minimization, and (e) cost-effective conservation, climate change mitigation, and fire risk reduction. Collectively the multiple objectives are hereafter referred to as the ‘multicriteria objective’.

2. Methods

Mathematical linear programming (Gurobi Optimizer 2021) was used to select for peatlands that yielded the highest ecosystem services benefits in Sumatra and Kalimantan for the period of 2015–2030. The period reflected the time from when the Government of Indonesia initiated national-level peatland restoration, to the designated end points of the UN Decade on Ecosystem Restoration and Indonesia’s Nationally Determined Contribution (NDC) towards climate change mitigation (Government of Indonesia 2016, United Nations General Assembly 2019). Sumatra and Kalimantan were selected for study as the regions accounted for 62%–75% of peatlands in Indonesia (Warren *et al* 2017b), more than 50% of which have been degraded or converted to industrial plantations and smallholder areas (Miettinen *et al* 2016a). Analyses were conducted across all priority provinces targeted by BRG for restoration (BRG 2017) (table S1 (available online at stacks.iop.org/ERL/17/064028/mmedia)), with the exception of Papua due to large uncertainty in estimation for carbon fluxes and storage (Warren *et al* 2017b) and missing data for canals and species distribution.

A 1 km² grid was overlaid on cartographic data of peatland extent (Wahyunto *et al* 2003, 2004), producing 2080 921 km² cells, or planning units. The resolution represents a tradeoff between processing load and the scale at which land use decisions are made. Where necessary, all data were resampled to 1 km² using nearest neighbor and bilinear interpolation for categorical and continuous data, respectively. The following were determined for each planning unit: (a) restoration suitability; (b) potential climate change mitigation benefits for 15 years; (c) fire risk reduction potential; (d) overall species extinction risk; and (e) economic costs for restoration (table 1). Areas suitable for restoration were determined using a 2015 land cover map (Miettinen *et al* 2016a), where settled and permanently wet and seasonally flooded areas were deemed unsuitable for restoration and removed. Pristine peat swamp and mangrove forests were also removed as areas not requiring restoration. The optimization algorithm was then run across increments of 10% of suitable peatlands for four different objectives (fire risk reduction, climate change mitigation, species conservation and cost minimization), to explore the extent to which varying the area restored impacted the costs and benefits of restoration. Additionally, the models were also run with BRG’s goal of restoring 25 000 km² of peatlands, as

the constraint. This allowed the results to be situated and discussed in the context of a real-world policy. For the carbon offset price, potential climate change mitigation was divided by the total cost of restoration.

The analyses were repeated with a second constraint limiting restoration to areas that would not affect current crop production level, assuming that the crop yield gap between current estimated productivity and the maximum that is feasible (75% of projected possible productivity) could be closed with improved technologies and management practices (Mueller *et al* 2012). Actual and potential yield gap data for seven rainfed tropical crop groups (Carrasco *et al* 2017) were obtained from IIASA/FAO (2012). Trade-offs and synergies of the objectives were explored further by combining species conservation, climate change mitigation potential, fire risk reduction potential, and restoration costs. Weightings were applied to the pairwise combinations of climate change mitigation potential, conservation, and fire risk reduction objectives at intervals of 20% to explore a range of scenarios for trade-offs and synergies (Strassburg *et al* 2020).

Finally, the optimization and trade-off analyses were repeated using groupings of 1 km² grid cells into the coarser analytical scales defined by village and catchment boundaries (figure S1). This resulted in a total of 2334 villages and 389 catchments. Further details regarding model setup and the uncertainty analysis are provided in Supplementary Information.

3. Results

3.1. Impacts of restoration

Peatlands in Riau were commonly prioritized for fire risk reduction and species conservation, those in South Sumatra for climate change mitigation, and those in Central Kalimantan, West Kalimantan, and Riau for low costs (figure 2). The majority of selected sites were located on anthropogenic land covers, such as large plantations for pulpwood and palm oil production, and smallholder farms (figure 3), which are also highly degraded. Forests were unlikely to be prioritized except for the cost minimization objective.

Peatland restoration contributed substantially to ecological and social wellbeing through reducing fire risk by 6%–37% (figure 4(a)). Maximum annual climate benefit was 0.36 Pg CO₂e yr^{−1}, or a total of 5.44 Pg CO₂e over 15 years, for the climate objective (figure 4(b)). This annual climate benefit amounted to approximately 26% of total emissions from the agriculture, forestry and other land use sectors in Indonesia in 2015 (Climate Watch 2021) (1.40 Pg CO₂e). Baseline extinction risk prior to restoration was approximately 14% summed across all species. For the conservation objective, extinction risk dropped to 0% after more than 20% of peatlands, or all the lost habitats between 2000 and 2015, were restored (figure 4(c)). At the minimum, restoration

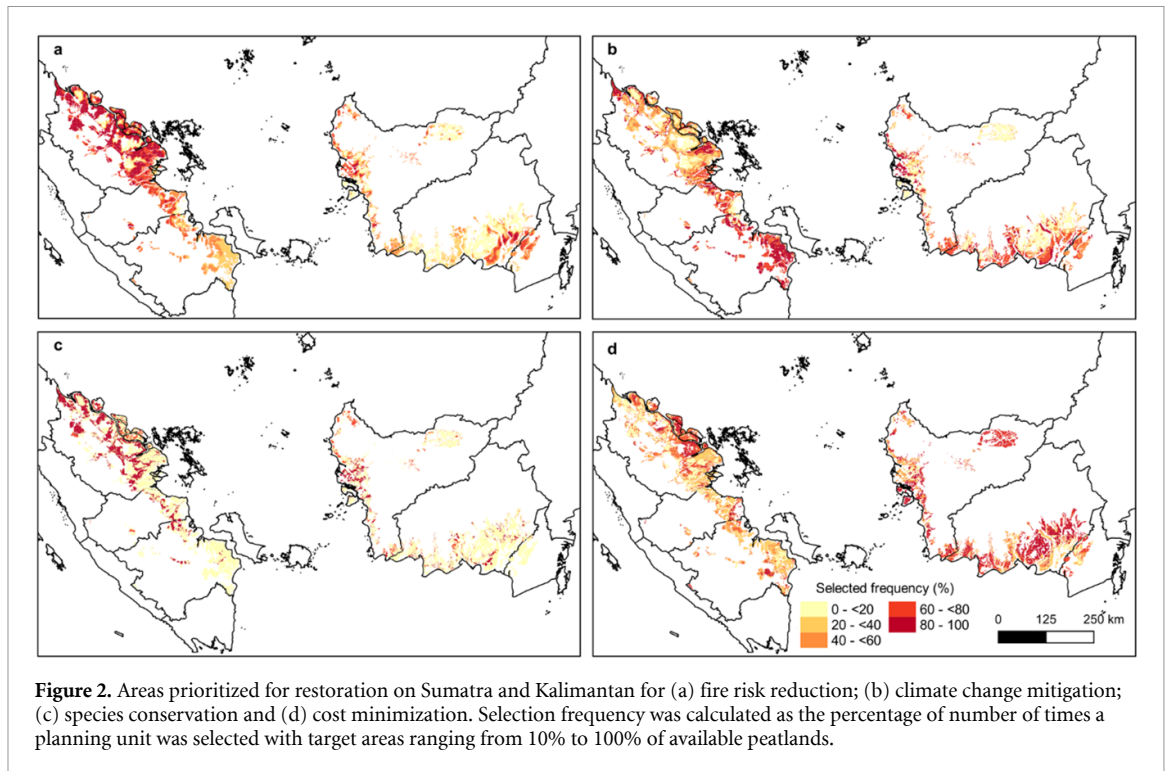
Table 1. Calculations for fire risk reduction, climate change mitigation, species conservation, and cost minimization objectives.

Objective	Calculation	Details
Fire risk reduction	Fire risk reduction = $\text{Fire}_{\text{post}} - \text{Fire}_{\text{pre}}$ where $\text{Fire}_{\text{post}}$ = fire risk post restoration (%) Fire_{pre} = fire risk in 2015, pre-restoration (%)	Fire risk reduction potential was predicted using a generalized linear mixed-effects model combining biophysical and social explanatory variables (table S2; Tan <i>et al</i> 2020). The year 2015 was used as a baseline, corresponding to the year of severe forest and land fires in Indonesia that sparked several peatland-related policies. A gain-loss approach was used to estimate climate change mitigation benefits derived from restoration (Budiharta <i>et al</i> 2018, Zeng <i>et al</i> 2020). Values for the variables in the equation were taken from literature (table S3) and where possible, assigned according to land use classes.
Climate change mitigation	$\Delta C = C_{\text{oxid}} + C_{\text{seq}} + C_{\text{burn}} - C_{\text{stock}} - C_{\text{CH}_4}$ where ΔC = Net carbon gain/loss C_{oxid} = Avoided emissions from peat oxidation C_{seq} = Carbon sequestration post-restoration C_{burn} = Avoided emissions from biomass fires C_{stock} = Initial carbon stock under land cover i C_{CH_4} = Methane emission post restoration expressed as CO ₂ e	Maps of the distribution of ten peat swamp forest-dependent bird species (table S4) for the years 2000 and 2015 (Symes <i>et al</i> 2018) were used to estimate the potential area of habitat to restore. Extinction risk was calculated using the equation by Strassburg <i>et al</i> (2020) and Neeson <i>et al</i> (2015).
Species conservation	$\sum_i^N B_i = S_{2000} - S_{2015}$ where B_i = Conservation benefit for planning unit i S_{2000} = Suitable habitat for n species in planning unit i for the year 2000 S_{2015} = Suitable habitat for n species in planning unit i for the year 2015 $\sum_i^N r = 1 - \left(\frac{A_{\text{current}}}{A_{\text{old}}} \right)^z$ where r = Extinction risk A_{current} = Restored habitat area and existing habitat area in 2015 A_{old} = Original habitat area; i.e. species distribution in 2000 z = Rate of diminishing return for reducing extinction risk despite additional habitat, set at 0.25 as per literature (Strassburg <i>et al</i> 2020)	Maps of the distribution of ten peat swamp forest-dependent bird species (table S4) for the years 2000 and 2015 (Symes <i>et al</i> 2018) were used to estimate the potential area of habitat to restore. Extinction risk was calculated using the equation by Strassburg <i>et al</i> (2020) and Neeson <i>et al</i> (2015).
Cost minimization	$\sum_i^N \text{cost} = D_{\text{rewet}} + D_{\text{revege}} + D_{\text{AR}} + D_{\text{fire}}$ where D_{rewet} = Cost of rewetting planning unit i D_{revege} = Cost of revegetating planning unit i D_{AR} = Agriculture rent in planning unit i ; proxy for the opportunity costs of giving up current land use D_{fire} = Cost of firefighting in planning unit i	The cost of restoration was set as the sum of four activities: rewetting, revegetation, revitalization (expressed as opportunity cost), and fire reduction (Dohong <i>et al</i> 2018). Values for the variables in the equation were taken from literature (table S5). For rewetting and revegetation, costs were stratified according to canal type (Hansson & Dargusch 2017) and forest degradation level (Budiharta <i>et al</i> 2014). The opportunity cost was derived from an agriculture rent map (Carrasco <i>et al</i> 2017) and firefighting cost was based on a fixed value derived from literature (Simorangkir 2007).

could cost as little as US \$7897–24 023 km⁻² yr⁻¹ depending on the area restored (figure 4(d)), but in return performed poorly for the other objectives. When crop production losses were constrained to be compensated with partial yield gap closure, restored area could not exceed 10%–40% of peatlands for groundnut, soybean, and wetland rice, and 60%–70% of peatlands for maize without compromising current

crop production levels within peatlands. As the actual production of cassava, oil palm, and sugarcane has already exceeded 75% of potential within peatlands, the models were unable to select sites for restoration without compromising current production levels.

Restoration efforts that met BRG's target of 25 000 km² could achieve relatively high fire risk reduction, climate change mitigation, and species



conservation benefits (figures S2 and S3). Selected sites were predominantly located on smallholder and industrial plantation land covers for all objectives except cost minimization, which prioritized degraded peat swamp forest. Up to 30% of average fire risk could be reduced and net emissions of $9992 \text{ Mg CO}_2\text{e km}^{-2} \text{ yr}^{-1}$ mitigated for the fire and climate objectives, respectively. The latter amounted to a total of $0.21 \text{ Pg CO}_2\text{e}$ sequestered and $3.54 \text{ Pg CO}_2\text{e}$ emissions avoided for 2015–2030. Even though $25\,000 \text{ km}^2$ is only approximately 25% of available peatlands, prioritizing certain species may reduce extinction risk to zero. The total cost to restore this area ranged from $\text{US } \$8912\text{--}38\,666 \text{ km}^{-2} \text{ yr}^{-1}$.

3.2. Multicriteria optimization

Combining objectives in a multicriteria optimization resulted in cost-efficient solutions at the expense of conservation, climate, and fire risk reduction benefits (figures S4–S6). Thus, when compared with their respective singular objectives at $25\,000 \text{ km}^2$, extinction risk reduction potential fell by 29%–71%, averaged annual climate benefit per area declined by 3%–80%, fire risk reduction potential was lowered by 24%–57%, whereas total costs increased by 16%–89%.

Areas that deliver maximum climate, conservation, and fire co-benefits were scattered throughout the provinces (figure S7). These areas tended not to overlap due to a trade-off between the objectives, as shown by the Pareto efficiency frontiers (figure 5), which show sets of optimal solutions where the outcome of an objective cannot be improved without reducing the outcome of the other objective(s). A weak trade-off is evident when restoring

beyond 80% of available peatlands for climate and conservation or fire risk reduction and conservation benefits, as shown by the flattening slope of the efficiency frontier.

3.3. Scales of restoration

Prioritizing restoration at the scale of villages and catchments, as opposed to a 1 km^2 grid square, generated similar patterns of site selection. At $25\,000 \text{ km}^2$, average annual climate benefit per area at the village and catchment scales was reduced by 21%–29% and fire reduction by 43% for their respective objectives compared with restoring at the 1 km^2 scale (figures S8–S21). Extinction risk reduction potential also decreased by 25%–30%, whereas the minimum total cost required to restore the same area increased by 21%–28% (table 2).

3.4. Economics of restoration

When the BRG's target of $25\,000 \text{ km}^2$ of peatlands was restored, cost per unit carbon at the 1 km^2 grid scale was priced at $\text{US } \$2\text{--}19/\text{Mg CO}_2\text{e}$ and at $\text{US } \$2\text{--}5/\text{Mg CO}_2\text{e}$ for the multicriteria objective. Whilst costs varied depending on the scale and area of restoration, results indicate that carbon prices are generally below $\text{US } \$37/\text{Mg CO}_2\text{e}$. The only exception was a carbon offset price of up to $\text{US } \$84$ at the catchment scale associated with the minimization of restoration cost objective. There was, however, a large uncertainty associated with the estimates due to varying costs for blocking canals of different sizes and carbon benefit estimations. For example, the maximum cost for blocking large, primary canals was estimated to be on average eight times that of small, tertiary canals (table S6).

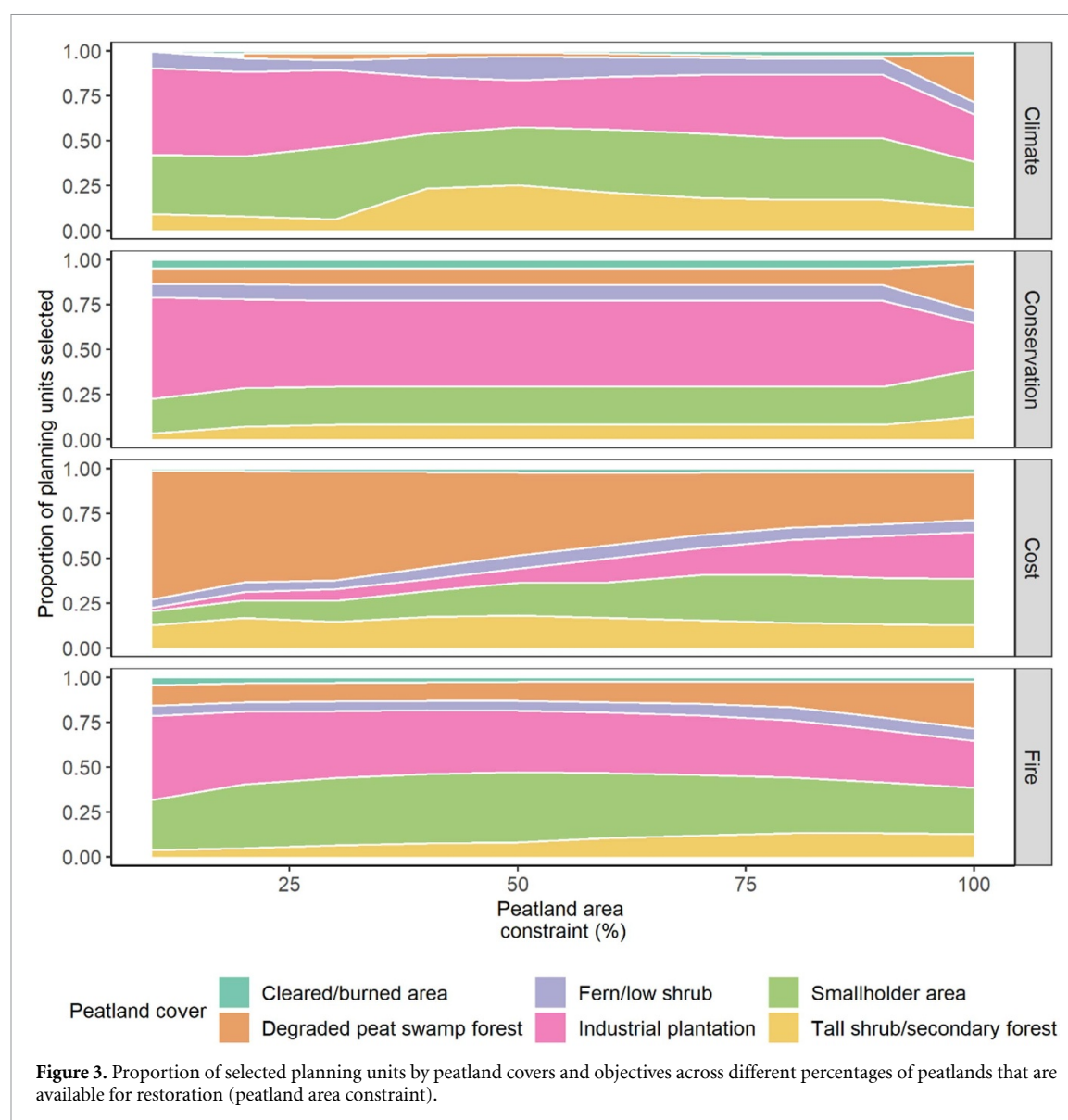


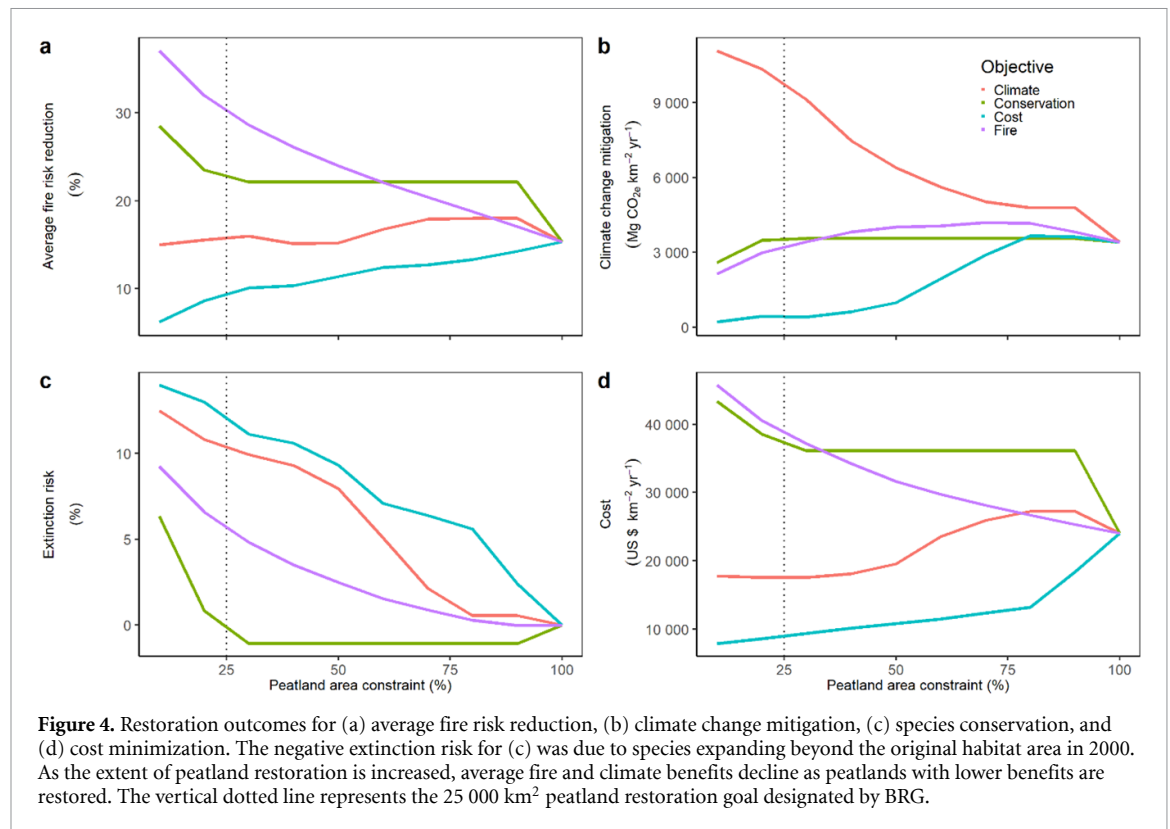
Figure 3. Proportion of selected planning units by peatland covers and objectives across different percentages of peatlands that are available for restoration (peatland area constraint).

4. Discussion

Restoring 25 000 km² of peatlands, the objective of the BRG in Indonesia, potentially reduces fire risk by up to 30% and also mitigates climate change by up to 0.25 Pg CO₂e yr⁻¹, offsetable by a highly affordable carbon price of USD 2–19/Mg CO₂e. Our analyses indicate that smallholders and industrial plantations are commonly prioritized across most objectives. These land covers are frequently drained and deforested, making them prone to fires, hence they hold the highest return benefit when restored. A previous study also reported prioritizing industrial plantation for fire mitigation (Santika *et al* 2020). Sites prioritized for fire risk reduction are located primarily in Riau, which in the recent past has been a major source of transboundary haze. Preventing peat fires can reduce atmospheric pollution in adjacent provinces and neighboring countries by an estimated 40%–70% and throughout tropical Asia by about 60% compared with a business-as-usual

scenario (Marlier *et al* 2015). Restoration and resultant reduced risk of haze-producing peatland fires could thus avoid substantial health losses across the region each year (Marlier *et al* 2019) and has important implications for climate change mitigation at the global scale.

In addition to fire reduction and associated protection of health, the restoration of tropical peatlands has the potential to enhance biodiversity conservation. By targeting only 25 000 km² of peatlands, restoration can recover lost habitats for endemic birds and presumably other associated taxa. However, how a restored peatland is managed is, ultimately, likely to determine its contribution to biodiversity conservation. Thus, Runting *et al* (2019) showed that improved management strategies were the optimum way of deriving both conservation and production benefits from forests in East Kalimantan province, and that with careful management both benefits could be extracted simultaneously.



Peatland restoration can potentially contribute to Indonesia's NDC by serving as a nature-based climate solution. NDCs are in effect national climate action plans that operationalize the 2015 Paris Agreement. The current research estimated a maximum climate benefit of 0.36 Pg CO₂e yr⁻¹ from carbon sequestration and avoided emissions when all peatlands are restored, comparable to estimates of 0.01–0.35 Pg CO₂e yr⁻¹ (Zeng *et al* 2020) and 0.50 Pg CO₂e yr⁻¹ (Griscom *et al* 2017) for restoring Southeast Asian and tropical peatlands, respectively. The extent of contribution is, however, constrained by the turnover time for restored peatlands to shift from carbon source to sinks. For example, sequestration benefit was substantially lower than avoided emissions. This is because peat swamp forest species are relatively slow-growing. As a result, the transition of a tropical peatland from a degraded carbon source to restored carbon sink can take many years (Taillardat *et al* 2020). An estimated 75 years is required, post-restoration, to recover 1/3 of the peat carbon lost as a result of the changes associated with the sustained plantation-based cultivation of oil palm (Warren *et al* 2017a). Carbon emissions from burning and oxidation are likely to decrease once all biomass has been depleted (Konecny *et al* 2016), however, and avoided emission benefit is likely to decline over time instead of remaining constant. There is thus a lag between restoration of a degraded peatland and carbon accumulation and eventual saturation that is not accounted for in the models, which represent an ideal, but highly improbable, scenario of immediate

restoration. Nevertheless, the long period for peatland ecosystems to become net carbon sinks again highlights the importance of protecting existing peat swamp forests and the need to consider including active management for enhanced carbon sequestration in restoration efforts (Taillardat *et al* 2020).

A trade-off between conservation or fire risk reduction and climate change mitigation receives support from other studies; climate and conservation benefits do not always overlap, despite being consistently lumped together as goals of land interventions (Phelps *et al* 2012, Budiharta *et al* 2014, Law *et al* 2015). Factoring in trade-offs therefore not only generates more realistic expectations for restoration, but also forces practitioners to explicitly account for diverse benefits. Adopting a multicriteria approach broadens the scope, allowing multiple ecosystem services and their interactions to be considered.

Difficult trade-offs are encountered at the intersection between maintaining livelihoods and peatland restoration. Models used in the current research prioritized smallholder and industrial plantations, or land covers that tend to be under active use. As such, there are limits to the extent of peatlands that can be restored without impeding levels of crop production after intensification. To better manage ecological benefits and economic functions, more transparent, scientifically-backed land use planning is needed to assign restoration and production functions. Mixed land use strategies combining restoration and agriculture can potentially reduce the magnitude of trade-offs (Law *et al* 2017). An increasingly popular method

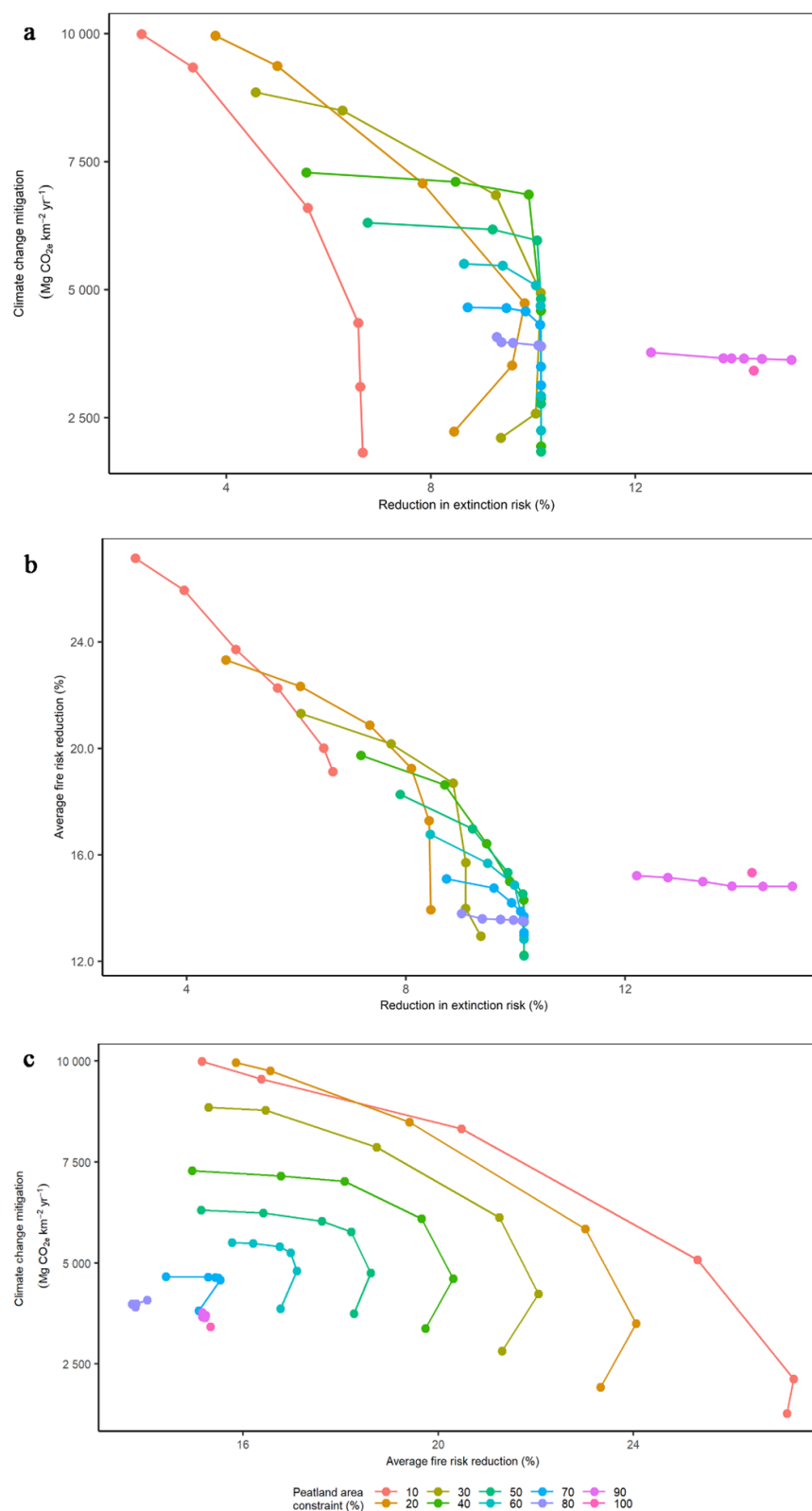


Figure 5. Optimal outcomes for the multicriteria objective, showing (a) climate and conservation benefits, (b) potential fire risk reduction and conservation benefits, and (c) potential fire risk reduction and climate benefits. Each point on the Pareto efficiency frontier represents an optimal scenario for which there is no better solution as an improvement in one objective comes at the cost of the other. Each connected group of points represents the same area constraint, but with different weights for either the conservation or climate objectives.

is paludiculture, or the production of biomass using native peatland species on rewetted peatlands (Tan *et al* 2021). Paludiculture can potentially achieve both ecological and economic objectives. However, the

long-term feasibility and sustainability of paludiculture systems remains debatable (Budiman *et al* 2020), while in the short-term paludiculture is unlikely to be adopted by farmers without compensation for

Table 2. Rounded outcomes of restoring 25 000 km² of peatlands at the 1 km², catchment, and village scales for their respective objectives.

Scale of analysis	Objective			
	Fire risk reduction (%)	Climate change mitigation (Pg CO ₂ e yr ⁻¹)	Extinction risk reduction (%)	Cost of restoration (US \$ CO ₂ e km ⁻² yr ⁻¹)
1 km ²	30	0.25	14	8912
Catchment	17	0.18	10	11 406
Village	17	0.20	11	10 803

reduced profits being made available (Schaafsma *et al* 2017). To this end, severe restrictions on the ability to close crop yield gaps to compensate production lost to restoration highlight the need to focus future agricultural intensification on existing agriculture lands.

Based on the respective objectives, increasing the geographical scales of analysis to the village and catchment level potentially reduced fire risks by 15%–18% and contributed 0.09–0.35 Pg CO₂e yr⁻¹ (6%–25% of the total emissions from the agriculture, forestry, and other land use sectors in 2015) to climate change mitigation. This contribution is comparable to the climate benefits obtained from restoring at 1 km². Despite their larger area, villages and catchments represent more ecologically- and socially-meaningful scales for capturing important ecological processes and distributing restoration resources. Moreover, whilst the 1 km² grid cell-level analyses produced optimal results, the coordination of restoration efforts and ability to derive benefits from economies of scale was rendered challenging by the often scattered and disconnected location of selected sites. Future optimization research exploring the effects of spatial scale and connectivity on restoration is needed to improve estimations of restoration costs and benefits.

Restoration costs could be partially offset through a market scheme, such as a system of carbon credits. A carbon price of US \$2–37/Mg CO₂e was needed to offset restoration cost across all objectives at the grid scale emphasizing the potential of peatland restoration as a cost-effective nature-based solution compared to alternative negative emission technologies. Higher carbon prices of US\$ 59–84/Mg CO₂e at the village and catchment scales reflect challenges with restoring across continuous landscapes that include areas with marginal carbon returns after restoration. Carbon credit schemes to pay for the reduction or removal of carbon emissions from the atmosphere via avoided emissions and sequestration are increasingly being promoted (Wei *et al* 2021). Depending on the estimation method, costs for removing carbon vary substantially for different negative emission technologies. For example, afforestation and reforestation are estimated to cost US \$10–237/Mg CO₂ based on studies utilizing spatial optimization models. By comparison, costs as low as US \$0.1–15/Mg CO₂ (Fuss *et al* 2018) have been established based on

estimations of various cost components. In general, restoration is a cheaper option compared with the other, more technological, negative emission technologies such as ocean fertilization or direct air capture (Fuss *et al* 2018, Taillardat *et al* 2020) and produce, as shown, diverse co-benefits. Aside from utilizing carbon markets as a source of funding, restoration costs could also be offset through the use of alternative market-based schemes, such as ecosystem restoration licenses (Harrison *et al* 2020).

5. Conclusion

There are several caveats in this study. An understanding of local conditions on the ground, such as land tenure, political governance, and land management methods is critical to decision-making regarding site allocation. Failure to consider, fully, local conditions could result in exclusion of communities, resistance from local actors and delay and even reverse restoration. However, because of the sensitive nature of fire use, often highly dynamic policy environment, and difficulties in acquiring reliable data of an appropriate scale, variations in local conditions are often insufficiently captured in optimization studies. This is particularly an issue when it comes to projecting future conditions. Future land use competition was not accounted for in the models used in this study, for example. Competition over land is likely to be impacted by the effects of future policies, including the Indonesian government's plans to expand the area of cultivated land in order to alleviate food insecurity concerns, with some of the expansion taking place on peatlands (SIIA 2021). Some areas designated under this program overlap with sites that are prioritized in our analyses for joint conservation and climate objectives. Social, legal, and economic factors therefore pose severe constraints to the realization of the restoration sites identified by the optimization models (Zeng *et al* 2020).

Spatiotemporal limitations in variables used in the models can also influence the estimates of restoration benefits and costs. For example, fire risk reduction was based on a dry El Niño year, despite differences in fire risks between dry and wet years (Tan *et al* 2020). The frequency and intensity of droughts and the risks of fire are likely to increase

due to climate change (Turetsky *et al* 2015), hence projected fire risk reduction in the current paper can be viewed as conservative. Given that fire reduction confers health benefits by mitigating toxic emissions, future optimization research focused on peatland restoration could include health outcomes, such as disability-adjusted life years, to capture both local and transboundary costs and benefits.

In light of these challenges, results presented here are perhaps best viewed as providing a means of highlighting the potential costs and benefits of peatland restoration in Indonesia. Decisions over the actual siting, planning and implementation of restoration sites should also include fine-grained information on local actors, politics, and environmental and socioeconomic conditions.

With large areas of degraded lands, restoration is needed more than ever to recover lost ecosystem services. Tropical peatland restoration is shown to be an affordable nature-based climate solution that generates climate change mitigation, conservation and health co-benefits in Indonesia. In addition to Indonesia, peatland restoration also features in the NDCs of several tropical countries, and forms part of the portfolios of many international climate and conservation initiatives. The current research found that the estimated cost of peatland restoration is less than other negative emission technologies, and thus has the potential to provide a highly cost-effective contribution to meeting sustainable development aims, particularly in less economically-advanced parts of the world. Nevertheless, the study indicates a weak trade-off between conservation or fire risk reduction and climate change mitigation objectives. By highlighting optimized co-benefits, the application of spatial optimization to rich datasets that include high quality information at the local scale is shown here to provide a means of prioritizing tropical peatland sites for effective restoration.

Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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Conflict of interest

The authors declare no competing interests.

ORCID iD

Z D Tan <https://orcid.org/0000-0001-5854-6987>

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